

Table 1: Experiment 8: SafeDreamer vs MACPO on MAMuJoCo (continuous actions).

	Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$
2x3HalfCheetah	5.0	MACPO Paper		10M	2250.0	40.0
2x3HalfCheetah	5.0	MACPO Run		10M	1316.8 ± 115.7	5.0 ± 1.7
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-5)		100k	8.5	0.0
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-5)		500k	1410.6	0.0
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-5)		700k	1455.9	15.0
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-5)		800k	1450.3	22.2
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-5)		900k	1685.9	37.8
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-5)		1M	1519.2	35.8
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-4)		100k	393.4 ± 466.5	0.0
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-4)		500k	1554.6 ± 631.6	0.8 ± 1.3
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-4)		700k	1801.6 ± 696.3	4.6 ± 4.1
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-4)		800k	1931.8 ± 820.7	127.0 ± 208.0
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-4)		900k	1744.9 ± 724.4	66.8 ± 100.4
2x3HalfCheetah	5.0	SafeDreamer (lr=1e-4)		1M	1480.8 ± 97.6	4.6 ± 2.0
2x3HalfCheetah	25.0	SafePO Paper		10M	1637.3	52.1
2x3HalfCheetah	25.0	MACPO Run		10M	1374.3 ± 85.7	22.0 ± 4.2
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-5)		100k	1136.2 ± 564.6	30.1 ± 30.2
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-5)		500k	2000.5 ± 120.6	5.4 ± 6.9
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-5)		700k	2587.8 ± 181.4	22.3 ± 22.2
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-5)		800k	2545.2 ± 212.5	37.6 ± 9.4
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-5)		900k	2528.5 ± 162.9	3.9 ± 3.7
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-5)		1M	2527.2 ± 228.6	16.6 ± 2.0
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-4)		100k	-60.1 ± 39.0	0.0
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-4)		500k	743.2 ± 408.4	0.0
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-4)		700k	969.0 ± 587.9	0.0
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-4)		800k	1019.7 ± 396.6	0.0
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-4)		900k	1070.2 ± 308.5	0.0
2x3HalfCheetah	25.0	SafeDreamer (lr=1e-4)		1M	1149.1 ± 312.7	0.0
2x4Ant	0.2	MACPO Paper		10M	900.0	15.0
2x4Ant	0.2	MACPO Run		10M	859.2 ± 32.7	1.2 ± 0.7
2x4Ant	0.2	SafeDreamer (lr=1e-5)		100k	894.0 ± 51.4	0.0
2x4Ant	0.2	SafeDreamer (lr=1e-5)		500k	1410.2 ± 394.7	2.0 ± 1.8
2x4Ant	0.2	SafeDreamer (lr=1e-5)		700k	1566.0 ± 857.2	18.0 ± 11.8
2x4Ant	0.2	SafeDreamer (lr=1e-5)		800k	1626.6 ± 158.9	43.5 ± 8.2
2x4Ant	0.2	SafeDreamer (lr=1e-5)		900k	1902.4 ± 359.2	43.9 ± 29.7
2x4Ant	0.2	SafeDreamer (lr=1e-5)		1M	2267.8 ± 134.1	24.3 ± 5.6
2x4Ant	0.2	SafeDreamer (lr=1e-4)		100k	903.4 ± 27.3	0.0
2x4Ant	0.2	SafeDreamer (lr=1e-4)		500k	1187.2 ± 137.5	0.0
2x4Ant	0.2	SafeDreamer (lr=1e-4)		700k	1184.0 ± 499.9	5.1 ± 5.0
2x4Ant	0.2	SafeDreamer (lr=1e-4)		800k	1373.5 ± 727.7	2.0 ± 2.8
2x4Ant	0.2	SafeDreamer (lr=1e-4)		900k	1494.8 ± 831.1	3.7 ± 6.0
2x4Ant	0.2	SafeDreamer (lr=1e-4)		1M	1618.4 ± 640.8	4.1 ± 5.8
2x4Ant	25.0	SafePO Paper		10M	1099.2	3.4
2x4Ant	25.0	MACPO Run		10M	798.1 ± 109.4	10.5 ± 1.7
2x4Ant	25.0	SafeDreamer (lr=1e-5)		100k	836.2 ± 93.2	0.0
2x4Ant	25.0	SafeDreamer (lr=1e-5)		500k	1086.0 ± 412.1	6.8 ± 11.5
2x4Ant	25.0	SafeDreamer (lr=1e-5)		700k	1613.5 ± 553.1	53.1 ± 47.1
2x4Ant	25.0	SafeDreamer (lr=1e-5)		800k	1794.4 ± 672.1	82.9 ± 72.0
2x4Ant	25.0	SafeDreamer (lr=1e-5)		900k	1838.3 ± 583.0	32.7 ± 33.7
2x4Ant	25.0	SafeDreamer (lr=1e-5)		1M	1835.8 ± 678.3	17.4 ± 15.1
2x4Ant	25.0	SafeDreamer (lr=1e-4)		100k	917.0 ± 58.0	0.0
2x4Ant	25.0	SafeDreamer (lr=1e-4)		500k	1207.0 ± 175.0	0.1 ± 0.2
2x4Ant	25.0	SafeDreamer (lr=1e-4)		700k	1156.7 ± 477.8	1.4 ± 1.2

Table 1 – continued

	Env	CL	Source	Steps	Reward $\uparrow$	Cost $\downarrow$
2x4Ant	25.0		SafeDreamer (lr=1e-4)	800k	1221.1 $\pm$ 478.7	2.0 $\pm$ 0.6
2x4Ant	25.0		SafeDreamer (lr=1e-4)	900k	1323.1 $\pm$ 629.2	7.0 $\pm$ 7.2
2x4Ant	25.0		SafeDreamer (lr=1e-4)	1M	1552.4 $\pm$ 475.2	22.6 $\pm$ 18.3
4x2Ant	1.0		MACPO Paper	10M	650.0	15.0
4x2Ant	1.0		MACPO Run	10M	1097.6 $\pm$ 97.4	3.4 $\pm$ 1.0
4x2Ant	1.0		SafeDreamer (lr=1e-5)	100k	916.3 $\pm$ 15.3	0.0
4x2Ant	1.0		SafeDreamer (lr=1e-5)	500k	965.2 $\pm$ 34.3	0.3 $\pm$ 0.6
4x2Ant	1.0		SafeDreamer (lr=1e-5)	700k	886.0 $\pm$ 110.8	0.0
4x2Ant	1.0		SafeDreamer (lr=1e-5)	800k	1134.0 $\pm$ 170.2	0.1 $\pm$ 0.1
4x2Ant	1.0		SafeDreamer (lr=1e-5)	900k	1140.6 $\pm$ 33.1	0.5 $\pm$ 0.8
4x2Ant	1.0		SafeDreamer (lr=1e-5)	1M	1241.0 $\pm$ 565.5	2.5 $\pm$ 3.1
4x2Ant	1.0		SafeDreamer (lr=1e-4)	100k	960.3 $\pm$ 33.3	0.0
4x2Ant	1.0		SafeDreamer (lr=1e-4)	500k	1069.5 $\pm$ 106.2	0.1 $\pm$ 0.1
4x2Ant	1.0		SafeDreamer (lr=1e-4)	700k	1432.8 $\pm$ 404.7	4.2 $\pm$ 7.3
4x2Ant	1.0		SafeDreamer (lr=1e-4)	800k	1378.4 $\pm$ 295.2	7.7 $\pm$ 13.4
4x2Ant	1.0		SafeDreamer (lr=1e-4)	900k	1595.7 $\pm$ 410.7	2.5 $\pm$ 4.0
4x2Ant	1.0		SafeDreamer (lr=1e-4)	1M	1558.8 $\pm$ 359.0	4.2 $\pm$ 7.3
4x2Ant	25.0		SafePO Paper	10M	815.8	0.0
4x2Ant	25.0		MACPO Run	10M	1287.3 $\pm$ 148.3	23.6 $\pm$ 6.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)	100k	902.4 $\pm$ 49.0	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)	500k	997.3 $\pm$ 83.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)	700k	1192.7 $\pm$ 86.6	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)	800k	1265.9 $\pm$ 101.8	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)	900k	1406.8 $\pm$ 172.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-5)	1M	1574.5 $\pm$ 304.2	0.5 $\pm$ 0.8
4x2Ant	25.0		SafeDreamer (lr=1e-4)	100k	918.7 $\pm$ 12.6	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-4)	500k	1025.5 $\pm$ 69.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-4)	700k	1168.3 $\pm$ 497.4	0.0
4x2Ant	25.0		SafeDreamer (lr=1e-4)	800k	1221.5 $\pm$ 220.3	0.1 $\pm$ 0.2
4x2Ant	25.0		SafeDreamer (lr=1e-4)	900k	1260.2 $\pm$ 302.6	0.7 $\pm$ 1.2
4x2Ant	25.0		SafeDreamer (lr=1e-4)	1M	1504.2 $\pm$ 520.0	2.9 $\pm$ 4.8